21/10/2026

Experiment: 2

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SEMESTER-MS2 September 2026

COURSE NAME: CSA4701 Deep Learning

2. Covariance, Eigenvalues and Eigenvectors Analysis

Scenario

A healthcare dataset contains patients' height, weight, and blood pressure. Calculate the covariance matrix to identify relationships among the features and visualize the correlations.

Algorithm

1. Start the program.
2. Import NumPy, Pandas, and Matplotlib libraries.
3. Create a healthcare dataset containing patients' height, weight, and blood pressure.
4. Convert the dataset into a Pandas DataFrame.
5. Calculate the covariance matrix using the dataset.
6. Calculate the correlation matrix to identify relationships among the features.
7. Compute the eigenvalues and eigenvectors of the covariance matrix using NumPy.
8. Display the covariance matrix, correlation matrix, eigenvalues, and eigenvectors.
9. Create a summary table showing the mean, minimum, maximum, and standard deviation of each feature.
10. Plot a correlation heatmap to visualize the relationships among height, weight, and blood pressure.
11. Plot a scatter plot to visualize the relationship between height and weight.
12. Display the plots with suitable titles, labels, and correlation values.
13. Stop the program.

Source code

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt
```

```
# 1. Create a healthcare dataset
data = {
   "Height (cm)": [150, 155, 160, 165, 170, 175, 180, 185],
   "Weight (kg)": [50, 55, 60, 65, 70, 75, 80, 85],
   "Blood Pressure": [110, 115, 118, 122, 125, 130, 135, 140]
}

df = pd.DataFrame(data)

print("HEALTHCARE DATASET")
print(df.to_string(index=False))

# 2. Calculate covariance matrix
cov_matrix = df.cov()

print("\nCOVARIANCE MATRIX")
print(cov_matrix.round(2))

# 3. Calculate correlation matrix
corr_matrix = df.corr()

print("\nCORRELATION MATRIX")
print(corr_matrix.round(3))

# 4. Calculate eigenvalues and eigenvectors
eigenvalues, eigenvectors = np.linalg.eigh(cov_matrix)

print("\nEIGENVALUES")
```

```
print(np.round(eigenvalues, 2))

print("\nEIGENVECTORS")
print(np.round(eigenvectors, 3))

# 5. Create a summary table
summary = pd.DataFrame({
    "Feature": df.columns,
    "Mean": df.mean().values,
    "Minimum": df.min().values,
    "Maximum": df.max().values,
    "Standard Deviation": df.std().values
})

print("\nSUMMARY TABLE")
print(summary.round(2).to_string(index=False))

# 6. Plot 1: Correlation heatmap
plt.figure(figsize=(7, 5))

plt.imshow(corr_matrix, cmap="coolwarm", vmin=-1, vmax=1)
plt.colorbar(label="Correlation")
plt.xticks(range(len(df.columns)), df.columns, rotation=20)
plt.yticks(range(len(df.columns)), df.columns)

for i in range(len(df.columns)):
    for j in range(len(df.columns)):
        plt.text(
            j, i, f"{corr_matrix.iloc[i, j]:.2f}",
```

```
            ha="center", va="center"
        )

    plt.title("Healthcare Feature Correlation Heatmap")
    plt.tight_layout()
    plt.show()

    # 7. Plot 2: Height vs Weight
    plt.figure(figsize=(7, 5))
    plt.scatter(df["Height (cm)"], df["Weight (kg)"])
    plt.xlabel("Height (cm)")
    plt.ylabel("Weight (kg)")
    plt.title("Relationship Between Height and Weight")
    plt.grid(True, linestyle="--", alpha=0.4)
    plt.tight_layout()
    plt.show()
```

Output

CORRELATION MATRIX

	Height (cm)	Weight (kg)	Blood Pressure
Height (cm)	1.000	1.000	0.997
Weight (kg)	1.000	1.000	0.997
Blood Pressure	0.997	0.997	1.000

EIGENVALUES

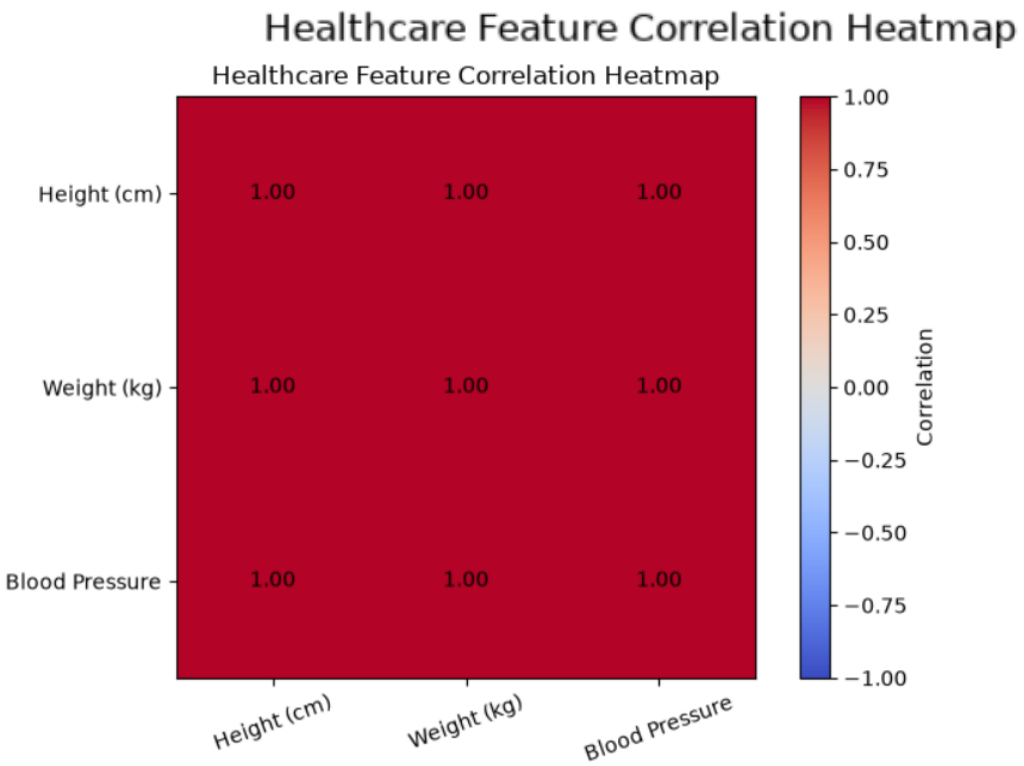
```
[  0.     0.52 403.75]
```

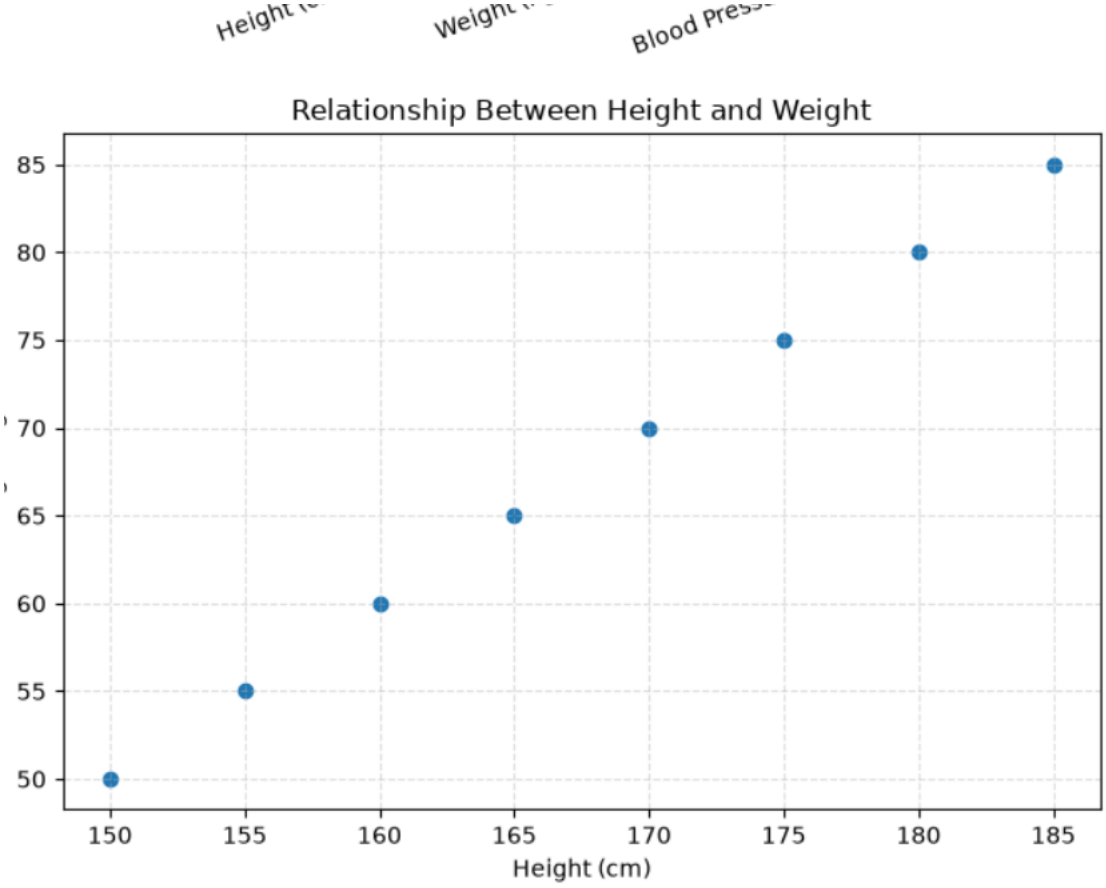
EIGENVECTORS

```
[[-0.707 -0.359 -0.609]
 [ 0.707 -0.359 -0.609]
 [ 0.     0.862 -0.507]]
```

SUMMARY TABLE

Feature	Mean	Minimum	Maximum	Standard Deviation
Height (cm)	167.50	150	185	12.25
Weight (kg)	67.50	50	85	12.25
Blood Pressure	124.38	110	140	10.21




Relationship Between Height and Weight
85
80
75
70
65
60
55
50
150
155
160
165
170
175
180
185
Height (cm)